vocal defenders of scientific realism. Maxwell also pointed out that the very idea of something that is "unobservable in principle" is subject to revision with new theories and new technology. Before the development of optics and the microscope, he pointed out, something "too small to be seen" would have been unobservable in principle. "It is theory, and thus science itself, which tells us what is or is not… observable," Maxwell wrote, echoing Einstein's words to Heisenberg. "There are no *a priori* or philosophical criteria for separating the observable from the unobservable."

With a better understanding of and appreciation for the history of science and the theory-laden practice of science, the realists also made quick work of positivist ideas about the functioning of science like instrumentalism and operationalism. The realists pointed out that if operational definitions are the ultimate definitions of scientific concepts, then there's no way to improve measuring processes or to design them in the first place, since that would require going beyond the operational definition. If, for example, length is defined as the thing that existing measuring sticks measure, then there's no way to design a better measuring stick, because we have a perfect one by definition. Yet scientists develop new and improved measurement devices all the time. The ideas of length, time, mass, and so on aren't merely defined by experimental operations: they are inherent in the theories used to design and test new measurement devices.

The realists' dismissal of operationalism was not fundamentally new—in fact, many positivists, including Carnap, had dismissed operationalism years earlier as too simplistic to account for the success and practice of science. But many positivists still clung to instrumentalism, the view that science is simply a tool for organizing and predicting perceptions, and that the metaphysical content of theories was unnecessary. The realists pointed out that this was untenable as well. If the unobservable "metaphysical" content of our best scientific theories—stuff like electrons—really bears no relation at all to the actual stuff in the world, then why do our scientific theories work at all? The theories themselves suggest explanations for the phenomena that we see based on the unobserved stuff. But if the unobserved stuff is just a convenient picture that just happens to come along with the "real" content of the theory (i.e., the predictions about the observable world) and that picture really doesn't line up with the stuff in the